Name: ____________________________ Date: ________________ Period: _________

Problems 1–6

1. The table below shows the number of daylight hours in a city on the first day of each month over one year.

Month x	1	2	3	4	5	6	7	8	9	10	11	12
Daylight $L(x)$ (hr)	9.0	10.2	11.8	13.3	14.5	15.0	14.5	13.3	11.8	10.2	9.0	8.5

(a) Identify the maximum and minimum daylight values and the months at which they occur.
Max = _________ at month ______; Min = _________ at month ______

(b) Compute $A = \dfrac{\max - \min}{2} =$ _________________ and $D = \dfrac{\max + \min}{2} =$ _________________

(c) Estimate the period k and compute B.
$k =$ _________ months $B = \dfrac{2\pi}{k} =$ _________________

(d) Using a cosine model with phase shift $C = x_{\max}$, write the model $L(x)$.
$L(x) =$ ____

(e) Use your model to predict the number of daylight hours at $x = 3.5$ (mid-April). ____________

2. A periodic function w has the following properties based on a data table: maximum value = 20, minimum value = 4, period = 10.

(a) Compute A and D. ____________

(b) Compute B. ____________

(c) If a maximum occurs at $x = 2$, write a cosine model $w(x)$. ____________

(d) Use your model to predict $w(7)$. ____________

3. A model for the height (in meters) of a point on a spinning amusement-park ride is $h(t) = 6\sin\left(\dfrac{\pi}{5}(t - 2.5)\right) + 8$, where t is time in seconds.

(a) Identify the amplitude, midline, period, and phase shift of h. ____________

____

(b) What is the maximum height? The minimum height? ____________

(c) At $t = 0$, the model gives $h(0) = 8 + 6\sin(-\pi/2) = 2$ m. Interpret this value in context. ____________

____

4. (AP-Style MC) A data set of periodic values has a maximum of 50 and a minimum of 10. Which of the following correctly gives the amplitude and midline of a sinusoidal model for this data?

(A) Amplitude = 50, midline = 10

(B) Amplitude = 40, midline = 30

(C) Amplitude = 20, midline = 30

(D) Amplitude = 30, midline = 20

(E) Amplitude = 20, midline = 40

Answer: _________

5. (AP-Style MC) The function $f(x) = 4\sin\left(\frac{\pi}{3}(x-1)\right) + 7$ models a periodic data set. Which of the following statements about f is correct?

(A) The period is $\frac{\pi}{3}$ and the maximum value is 11.

(B) The period is 6 and the maximum value is 11.

(C) The period is 6 and the maximum value is 7.

(D) The period is 3 and the maximum value is 11.

(E) The period is 2π and the maximum value is 11.

Answer: _________

Extension optional

Extension (Optional): Enter the daylight-hours data from Problem 1 into Desmos (desmos.com) and perform a sinusoidal regression by fitting $y = a\sin(bx + c) + d$ to the data.

(a) Record the regression equation: ______________________________

(b) Compare the regression values of a (amplitude) and d (vertical shift) to your hand-computed values from Problem 1. How close are they? What might explain any differences? ____________________

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Preview — Lesson 3.8: The Tangent Function
Sine and cosine model bounded, wave-like phenomena that oscillate between a maximum and minimum. In Lesson 3.8 we'll explore the *tangent* function — a periodic function that has no maximum or minimum and shoots off to $\pm\infty$ in every period. Come ready to connect tangent to the unit circle ratio $\tan\theta = \sin\theta / \cos\theta$.